

Dielectric Dispersion of Colloidal Suspensions in the Presence of Stern Layer Conductance: Particle Size Effects

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In this work, we discuss the role that particle size plays in the manifestations of surface conduction on the dielectric response of colloidal dispersions. To that aim, experimental data on the dielectric constant of polystyrene suspensions of two different particle diameters (23 and 530 nm) are first compared to the predictions of a classical or standard model (E. H. B. DeLacey and L. R. White, *J. Chem. Soc. Faraday Trans. 2* **77**, 2007 (1983)), and it is found that, while the latter explains reasonably the dielectric behavior of the smallest particles, it considerably underestimates the phenomenon in the case of large particles. To explain these results in terms of contributions of ion motions in the inner region of the double layer of the particles, the approach followed by C. S. Mangelsdorf and L. R. White (*J. Chem. Soc. Faraday Trans. 86*, 2859 (1990)) is used to incorporate surface conductance in the theory of dielectric response of suspensions. In ac fields it is found that the model considerably improves the comparison between theory and experiment, whereas its use seems unnecessary for the smallest particles, where, whatever the combination used for the parameters of the theory, its predictions do not differ from the standard theory. Only in the case of the larger particles studied does the introduction of surface conductance play any role. A comparison between both types of theoretical results in a wide range of particle sizes demonstrates that Stern layer conductance always increases the magnitude of the low-frequency dielectric constant of suspensions, but its effect is less important the smaller the particle size and the larger the zeta potential for fixed ionic conditions in the dispersion medium. © 1999 Academic Press

Key Words: low-frequency dielectric dispersion; dynamic Stern layer; colloidal suspensions.

INTRODUCTION

When an ac electric field $E = E_0 \exp(i\omega t)$ is applied to a dispersion of colloidal particles, the polarization induced in the double layer of the particles is, in general, out of phase with the applied field. As a consequence, a characteristic frequency dependence is shown by the real (ϵ'_r) and imaginary (ϵ''_r) parts of the complex dielectric constant of the suspension (1, 2):

$$\epsilon_r^*(\omega) = \epsilon'_r(\omega) - i\epsilon''_r(\omega). \quad [1]$$

Furthermore, $\epsilon'_r(\omega = 0)$ can be much larger than that of the dispersion medium.

If the particle volume fraction ϕ is low (dilute suspension), a linear relationship exists between ϵ_r^* and ϕ ,

$$\epsilon'_r(\omega) = \epsilon'_{rs} + \phi \Delta\epsilon'_r(\omega) \quad [2]$$

$$\epsilon''_r(\omega) = \phi \Delta\epsilon''_r(\omega), \quad [3]$$

where $\Delta\epsilon'_r(\omega)$ and $\Delta\epsilon''_r(\omega)$ are, respectively, the real and imaginary parts of the complex dielectric increment, $\Delta\epsilon_r^*(\omega)$, of the suspension. The latter quantity measures the contribution of the particles and their double layers to the overall dielectric behavior of the suspension. Typically, $\Delta\epsilon_r^*(\omega)$ presents a relaxation frequency in the kHz to MHz region; since other relaxations appear at higher frequencies (2), we speak of low-frequency dielectric dispersion (LFDD) to describe the frequency dependence of $\Delta\epsilon'_r(\omega)$ or $\Delta\epsilon''_r(\omega)$ in the frequency region where double layer polarization phenomena manifest.

In fact, the frequency spectrum of $\Delta\epsilon_r^*$ is intimately related to the properties of the particles (size, shape, and dielectric constant), to the properties of the ionic dispersion medium (concentration, valence, and mobility of each ionic species present), and, most important, to the characteristics of the solid/liquid interface. This is one of the reasons explaining the growing interest in LFDD as a method of electrokinetic characterization.

Like in any other electrokinetic phenomenon, a theoretical model is needed to relate the above-mentioned properties to the measured dielectric increment. A number of theories have been proposed since the pioneering work of Dukhin and Shilov (2), differing from each other in the starting assumptions. For instance, some authors propose that only a diffuse double layer contributes to polarization, the inner layer ions being immobile (1–6), while others assume that Stern layer ions can undergo electric migration, and diffusion, thus playing a role in LFDD (7–15), with either free or restricted ionic exchange between diffuse and Stern regions. Theories contemplating only diffuse layer polarization are sometimes considered as part of the so-called *classical* or *standard* model of dielectric dispersion.

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Among them, we will be concerned here mainly with the treatment elaborated by DeLacey and White (4) that, although requiring numerical integration, is applicable whatever the values of κa (κ is the reciprocal Debye length, and a is the radius of the spherical particles of the suspension), zeta potential (ζ), and concentrations, valencies, and mobilities of the ions in solution.

Rosen *et al.* (13) generalized the DeLacey and White (DW hereafter) model by incorporating Stern layer conductance to the explanation of LFDD, whereas Mangelsdorf and White (16), using a slightly different approach, calculated the electrophoretic mobility and DC conductivity of suspensions when ions in the Stern layer are allowed to move under the action of the applied field.

In this paper, we use the original approach of Mangelsdorf and White together with the DW model to include a dynamic Stern layer (DSL). The results are compared to experimental $\Delta\epsilon'$ values obtained with suspensions of spherical particles of very different sizes to discuss under which conditions the general DSL theory must be used or under which conditions the simpler DW model is sufficient to explain the LFDD data.

EXPERIMENTAL

The colloidal dispersions studied were obtained from polystyrene spheres supplied by IDC Latex Corporation (USA). Two latexes were used, and their characteristics are given in Table 1. The original dispersions were first thoroughly cleaned by serum replacement in a stirred filtration cell, using Milli-Q water. Subsequently, fractions of these *clean* latexes were subjected to the same process with KCl solutions of the desired ionic strength, and, finally, the volume fraction (0.01 for sample L-23 and 0.02 for sample L-530) was adjusted by addition of suitable amounts of solution.

Electrophoretic mobilities (μ_e) of the two latexes in the presence of KCl solutions were measured in very dilute suspensions (volume fractions around 10^{-4}) using a Malvern Zetasizer 2c (England) at $25.0 \pm 0.5^\circ\text{C}$. At least 10 independent determinations were performed for each system, and their standard deviation will be used as an error parameter.

The complex conductivity and dielectric response of the polystyrene suspensions were obtained by measuring (in the 1 kHz–1 MHz frequency range) the complex admittance of a conductivity cell with variable interelectrode distance with a Hewlett–Packard HP-4284A LCR meter (USA). The details of

TABLE 1
General Characteristics of the Polystyrene Latexes
Used in This Work

Sample	Average diameter (nm)	Surface charge density ($\mu\text{C}/\text{cm}^2$)
L-23	23 ± 3	1.2
L-530	530 ± 30	3.9

TABLE 2
Electrophoretic Mobilities (μ_e) and Zeta Potentials (ζ) of the
Polystyrene Latexes for the Electrolyte Concentrations Indicated

Latex	[KCl] (mM)	μ_e ($\mu\text{m s}^{-1}/\text{V cm}^{-1}$)	ζ^a (mV)	κa^b
L-23	0.1	-5.1 ± 0.8	-119	0.38
	0.5	-5.3 ± 0.4	-201	0.85
L-530	0.1	-3.23 ± 0.07	-56.3	8.7
	1	-4.65 ± 0.06	-72.75	27.5

^a Zeta potential deduced from μ_e using O'Brien and White's theory (20).

^b κ^{-1} is the double layer thickness, and a the particle radius.

the method used and its ability to reduce the undesired effects of electrode polarization can be found in Refs. (17–19). The conductivity cell was thermostatted at $25.0 \pm 0.1^\circ\text{C}$ by means of a Haake F3-K (Germany) thermostatic bath.

RESULTS AND DISCUSSION

Electrophoretic Mobility

A preliminary step in the electrokinetic characterization of the particles was to measure their electrophoretic mobility as a function of electrolyte concentration (KCl). Table 2 shows the results, together with the ζ potentials calculated from μ_e data according to O'Brien and White's theory (20). The absolute values of both the electrophoretic mobility and the ζ potential display a maximum when the concentration of KCl approaches 1 mM (sample L-530) and 0.5 mM (L-23). This behavior of ζ is unjustified from the point of view of the standard electrokinetic model, according to which increasing the concentration of an indifferent electrolyte can only compress the diffuse layer, thus reducing $|\zeta|$. In any case, this anomaly has been observed in many polystyrene latexes (21–23) and among the reasons that were offered to explain it was anomalous (i.e., Stern layer) surface conductance. It is expected that our further investigation on the dielectric behavior of the two latex systems will give us additional clues on the dynamics of the electric double layer of these colloidal particles.

Dielectric Dispersion of the Suspensions

Figure 1 shows the real part of the dielectric increment as a function of the frequency of the applied field for both samples (L-530 and L-23, see Table 1) and the KCl concentrations used. The dielectric relaxation is clearly observable in latex L-530, whereas the phenomenon is only suggested by the data obtained with the small particles, due to the fact that the maximum frequency available in the HP-4284A is 1 MHz and the characteristic frequency for L-23 particles is much higher. Nevertheless, in both cases a quantitative estimation of the main parameters of the relaxation curve (i.e., the low-frequency value of $\Delta\epsilon'$, also named dielectric amplitude, and the relaxation frequency, ω_{rel}) can be obtained by fitting the ex-

perimental data to the following relaxation pattern, proposed by Vogel and Pauly (24) and that has been found to describe suitably the relaxation behavior of $\Delta\epsilon'_r(\omega)$ in suspensions,

$$\Delta\epsilon'_r(\omega) = \Delta\epsilon'_r(\infty) + \frac{\Delta\epsilon'_r(0) - \Delta\epsilon'_r(\infty)}{(1 + \omega\tau)(1 + \sqrt{\omega\tau})}, \quad [4]$$

where $\tau (=1/\omega_{rel})$ is the characteristic time of the system, $\Delta\epsilon'_r(\infty)$ is the high-frequency dielectric increment, and $\Delta\epsilon'_r(0)$ is the relaxation amplitude mentioned above.

The results of these fittings are shown in Table 3. As expected, smaller particles mean shorter times for ion to flow back and forth around the particle under the action of the alternating field. Furthermore, $\Delta\epsilon'_r(0)$ is more significant for large particles (in fact, for large κa values) because of the larger phase lags between diffusive ionic flows and the applied field giving rise to more intense displacement currents, ulti-

TABLE 3
Parameters of Eq. [4] (See Text) for the Experimental $\Delta\epsilon'_r$ Values Shown in Fig. 1

[KCl] (mM)	Latex L-23		Latex L-530	
	$\Delta\epsilon'_r(0)$	$\tau (10^{-8} \text{ s})$	$\Delta\epsilon'_r(0)$	$\tau (10^{-8} \text{ s})$
0.1	2400	1.96	8300	1210
0.5	2520	2.69	—	—
1	—	—	25000	2950

mately responsible for the high observed values of $\Delta\epsilon'_r(\omega \rightarrow 0)$ (2, 25).

Comparison between Experimental Data and Predictions of the Standard Model

We now explore to what extent the results just shown are justified in the light of the standard model in the DW approach (4). Qualitative arguments given above confirm that the general trends of the results (in particular, larger values of $\Delta\epsilon'_r(0)$ and smaller characteristic times for L-530 as compared to L-23) are in agreement with the standard approach. To carry out a more quantitative comparison, we have calculated the DW predictions for $\Delta\epsilon'_r(\omega)$ pertaining to experimental data in Fig. 1, using the ζ potentials of Table 2. The results are displayed in Fig. 2: as observed, the classical explanation of the polarization of electric double layers significantly underestimates the dielectric increments found experimentally in the case of L-530 particles. Unlike the large particles, the dielectric constant of L-23 suspensions is reasonably described by the DeLacey and White's theory, despite the fact that it can be expected that the surfaces of both types of lattices will be quite similar, except for the presence of surfactant on the surface of L-23, which is necessary to produce stable suspensions with such small size.

Before proceeding with plausible explanations for these different behaviors, it is worth considering the possibility that particle–particle interactions might account for deviations from the standard model, one of whose hypotheses is that the observed dielectric response is just the sum of the contributions of individual particles. This can be qualitatively verified by comparing the average particle–particle distance (r_p) with the characteristic diffusion length of counterions in the polarized double layer (L_D). According to different authors (2, 25), L_D is typically of the order of the particle radius, so the ratio r_p/L_D for our systems will be ~ 15 (sample L-23) and ~ 10 (sample L-530). This means that the suspensions considered are dilute enough for the classical models to be applicable.

We try to analyze how their low-frequency dielectric dispersion can be managed by a more complete theory based on the simultaneous contributions of both diffuse and Stern layers to the overall polarization of the particle's ionic atmosphere (13, 16).

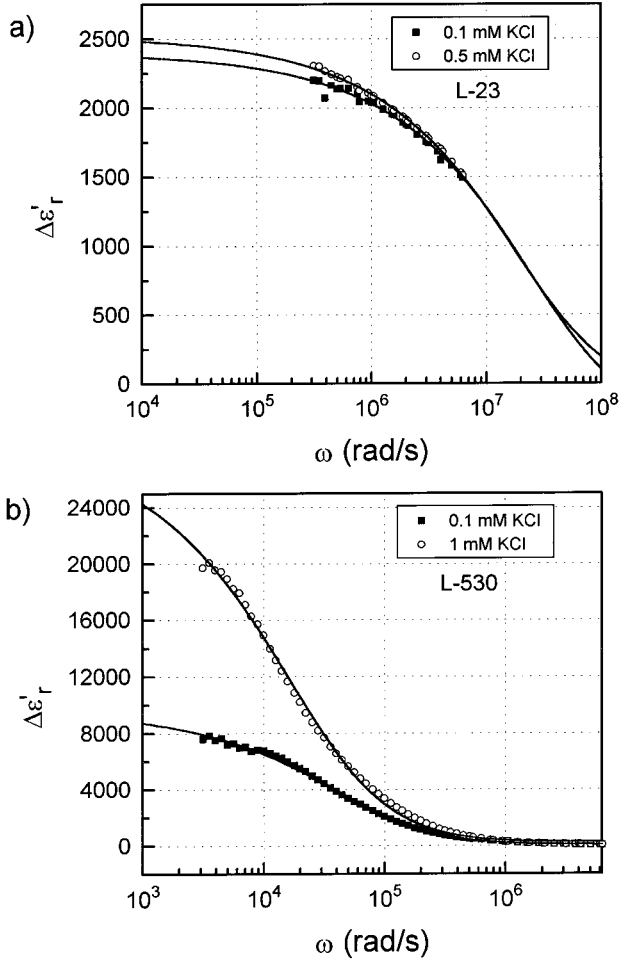


FIG. 1. Experimental values of the real part of the dielectric increment, $\Delta\epsilon'_r(\omega)$, as a function of frequency for latexes L-23 and L-530 and the concentrations of KCl indicated. Continuous lines: best fitting to Vogel and Pauly equation (cf. eq. [4]).

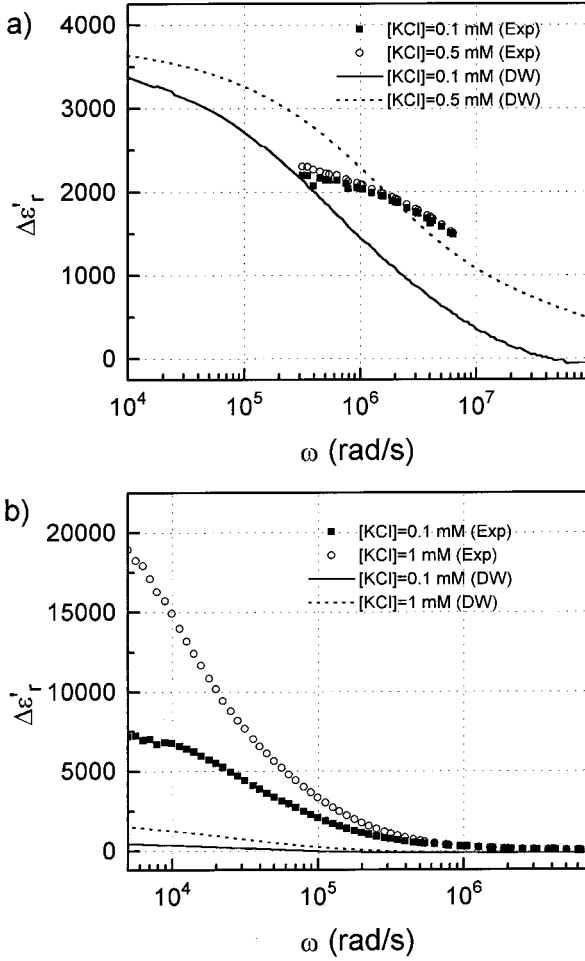


FIG. 2. Comparison between experimental (symbols) and predicted (lines) values of $\Delta\epsilon'_r$ for latexes L-23 (a) and L-530 (b) for the KCl concentrations indicated. Theoretical data were obtained with the DW model.

Comparison with DSL Model

According to Mangelsdorf and White (16), the lateral motion of Stern layer ions for a given external dc field depends on (i) the ζ potential (the shear plane is assumed to coincide with the beginning of the diffuse layer); (ii) the ratio between the drug coefficient of i th ionic species in the bulk solution (λ_i) and in the Stern layer (λ_i^s); (iii) the number N_i of sites available in the layer for adsorption of ions of type i ; (iv) the dissociation constant, K_i , of those sites; and (v) the capacity, C_2 , of the outer Stern layer. The effect of all these quantities is represented by a surface conductance parameter, δ_i , given by (16)

$$\delta_i = \frac{\frac{N_i}{K_i a} \frac{\lambda_i}{\lambda_i^s(\beta_1)} \exp\left(\frac{z_i e}{k_B T} \frac{\sigma_d}{C_2}\right)}{1 + \sum_{j=1}^N \frac{n_j^0(\infty)}{K_j} \exp\left[\frac{-z_j e}{k_B T} \left(\zeta - \frac{\sigma_d}{C_2}\right)\right]} \quad [5]$$

for each ionic species. In the case of ac fields, if the dissociation reactions in the Stern layer are considered fast enough, the reasonable hypothesis is made that both the overall surface charge σ_o and the Stern layer charge density σ_s are independent of time (26). The same can be said about the surface conductance parameters δ_i . In fact, Mangelsdorf and White (27, 28) have recently considered a general treatment in which both charge densities are allowed to change with time in the presence of an alternating field. These authors have demonstrated that the effect of time-varying surface charges on either the mobility or the dipole coefficient of the colloidal particle is negligible when compared to the DC surface conductance effect. Hence, in the present work, δ_i is considered constant, and stationary values are assumed for σ_o and σ_s , as in Refs. (16, 26).

Assuming that the supporting solution contains only two types of ions, and considering the predominant effect of counterions, we show only the results when δ_- changes (δ_- is the value of δ_i for counterions for a positively charged particle). Figure 3 demonstrates that, in agreement with results obtained by Rosen *et al.* (13), lateral motion of inner-layer ions is always favorable to increasing the dielectric constant of the suspensions. Furthermore, the effect can be extremely important, because the dielectric amplitude can increase by a factor of 10 for high (but physically reasonable) δ_- values.

To apply the model to our experimental data, the counterion parameters and ζ potential were changed until the differences between measured and predicted values of both $\Delta\epsilon'_r(\omega = 0)$ and μ_e were simultaneously minimized. Table 4 summarizes the

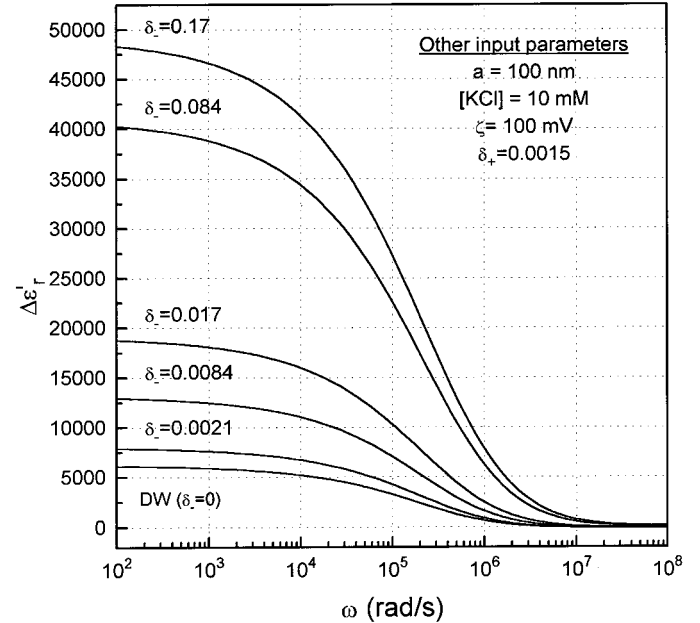


FIG. 3. Dielectric increment for suspensions of spherical particles, calculated with the Mangelsdorf-White DSL model for different values of δ_- , the surface conductance parameter of counterions. The DW predictions are included for comparison.

TABLE 4
Stern-Layer Conductance Parameters for Sample L-530
(Mangelsdorf–White Model, See Ref. (16))

	[KCl] = 0.1 mM	[KCl] = 1 mM
ζ (mV)	-130	-140
pK_+	2	2
eN_+ ($\mu\text{C}/\text{cm}^2$)	8	8
λ_+/λ_+^t	1	0.6
μ_e (predicted)	1.046	0.933
μ_e (experimental)		
$\Delta\epsilon_r'(0)$ (predicted)	0.87	1.016
$\Delta\epsilon_r'(0)$ (experimental)		

Note. Predicted values of μ_e and $\Delta\epsilon_r'(0)$ are also included. The following parameters were kept fixed: $eN = 4 \mu\text{C}/\text{cm}^2$; $\lambda_-/\lambda_-^t = 0.5$; $pK_- = 2$; Stern-layer capacitance = $130 \mu\text{F}/\text{cm}^2$.

results for samples L-530: note that ζ changes by less than 10% between $[\text{KCl}] = 0.1 \text{ mM}$ and $[\text{KCl}] = 1 \text{ mM}$ (compare with the change of $\sim 25\%$ if the standard model is used) and that the only parameter that has to be changed is the counterion drag ratio, λ_+/λ_+^t . According to these data, saturation effects cause the lateral ionic motion in the Stern layer to be somewhat hindered when the number of ions in the double layer is increased, which does not seem unreasonable. Figure 4 shows the full $\Delta\epsilon_r'(\omega)$ calculated curves together with experimental LFDD data; as observed, the agreement is considerably improved compared to classical DW predictions, except that the relaxation frequency is not well reproduced by DSL predictions.

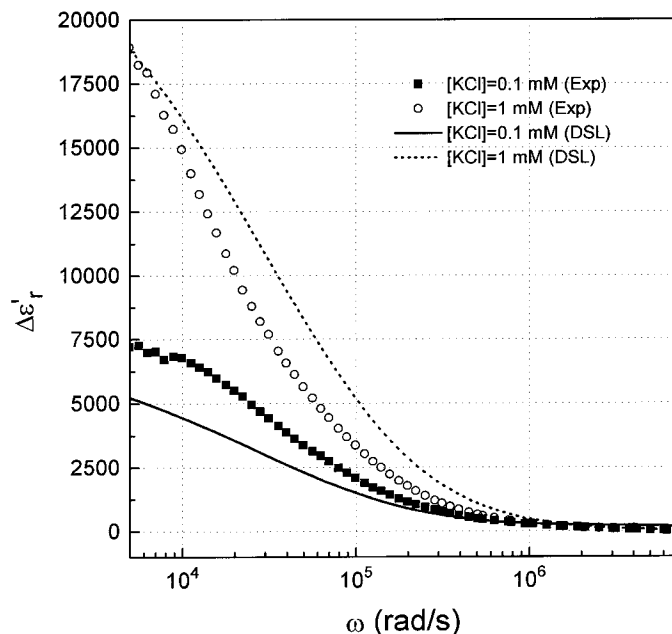


FIG. 4. Comparison between experimental (symbols) and predicted (lines) values of $\Delta\epsilon_r'$ for latex L-530 for the KCl concentrations indicated. Theoretical data were obtained with the DSL model, using the parameters in Table 4.

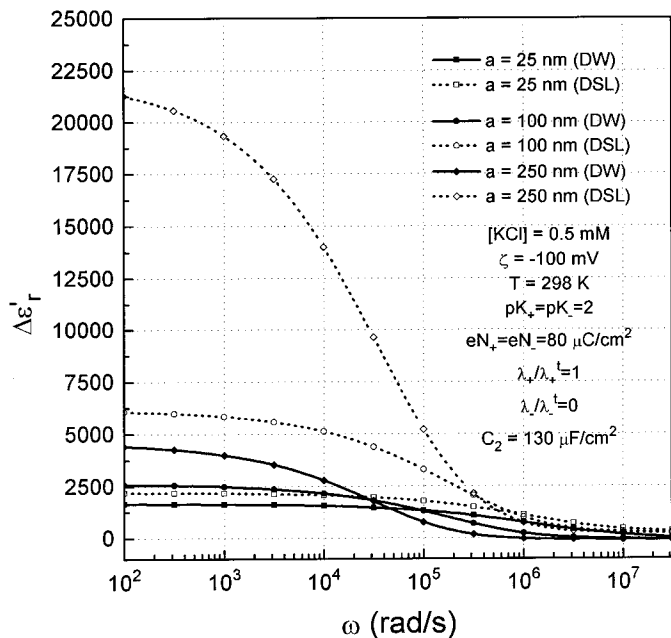


FIG. 5. Comparison between the frequency dependencies of $\Delta\epsilon_r'(\omega)$ calculated with the standard (DW) and DSL models, for different particle radii.

We repeated the same fitting process for L-23 particles. Whatever the changes introduced in the input parameters within reasonable physical ranges, we could never find significant differences between classical and DSL predictions. Only the ζ potential appeared to be the crucial quantity capable of modifying the model predictions on either the mobility or the dielectric increment. This suggests that DSL corrections are not needed if small particles (or, more correctly, small κa values) are used. Such a possibility has been further explored in Fig. 5, where the predictions of classical and DSL theories are compared for the frequency range of interest and three particle sizes. It is clearly seen that the low-frequency differences are negligible for small particles and become progressively more significant as the particle size is increased.

In Fig. 6 we show that, furthermore, if the ζ potential is reduced, the differences between DSL and DW approaches (that have been plotted as a function of particle size) increase. A qualitative explanation for the trends of variation of $(\Delta\epsilon_r'(0))_{\text{DSL}} - \Delta\epsilon_r'(0)_{\text{DW}}$ as a function of both ζ and particle radius a can be offered. Let us first consider the size effect: low particle radius (for any given ionic strength) means high double layer thickness (compared to a), that is, a very wide diffuse cloud. It is reasonable to assume that whatever the ionic conditions in the very thin Stern region (actually, it is considered two-dimensional in the DSL models), the role of the diffuse layer dipole will prevail over that of the inner layer. However, when the thickness of the former decreases (large κa), the influence of the latter will be enhanced, and the Stern layer effects may even (as shown in Figs. 5 and 6) largely exceed those generated in the diffuse layer. Concerning the ζ

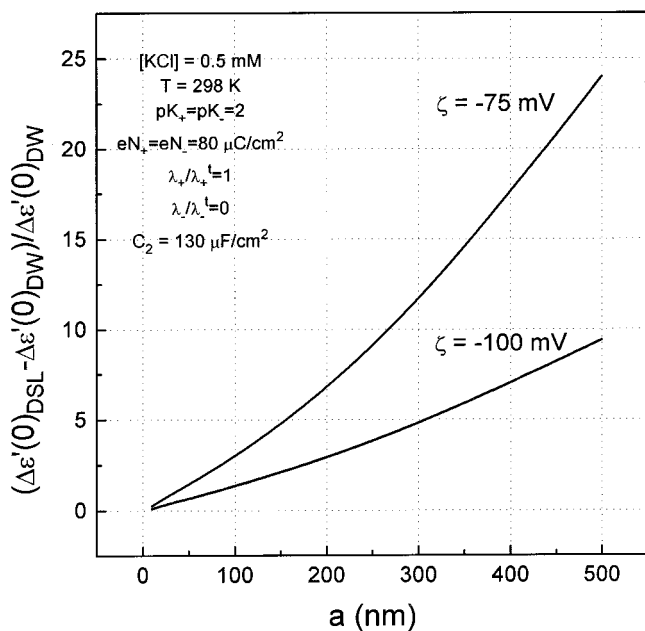


FIG. 6. Relative difference between the low-frequency dielectric increments predicted by DW and DSL theories, plotted as a function of particle radius, a , for two ζ potentials.

potential, a large ζ value will induce high counterion concentrations in the diffuse region of the double layer, thus hiding to some extent the polarization of Stern layer, less sensitive to ζ potential (N_i , λ_i^t , K_i are independent parameters, not directly linked to the value of ζ).

CONCLUSIONS

We have shown that, as previously reported for other electrokinetic phenomena, some discrepancies between theory and experiment can be explained by the existence of Stern layer surface conductance, that is, by using models (DSL) in which the inner layer ions can undergo tangential transport under the action of external fields. This is more likely for suspensions with large particles and/or high ionic strength ($\kappa a \gg 1$), since for low κa effects associated with polarization of the Stern layer are obscured by those of the thick diffuse layer. As a consequence, discrepancies found in the latter conditions are not likely to be justifiable under the assumption of Stern layer conductance alone.

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